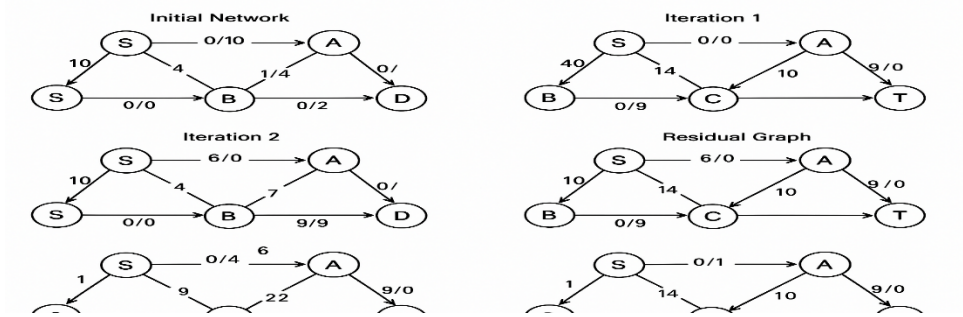
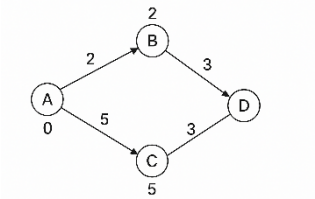
DEPARTMENT OF INFORMATION SCIENCE & ENGINEERING

Date: 17/12/2025	**CIE - II**	**Max. Marks: 60**
Semester: V	**UG**	**Duration: 2 Hour**
Course Title: ADVANCED ALGORITHMS		**Course Code: CS355TBB**

SCHEME AND SOLUTION

Sl. No.	PART A	M	BT	CO
1.	A graph showing remaining capacities.	1	1	1
2.	O(E·fmax), where fmax is the maximum flow value.	1	2	2
3.	Maximum matching → no more edges can be added (locally optimal). Maximum cardinality matching → largest number of edges (globally optimal)	2	2	3
4.	$\|V\|-1$	1	2	1
5.	Maximum flow	1	1	2
6.	Network Routing and Task Scheduling	2	3	3
7.	Efficient Time Complexity and Optimized for Reweighted Edges	2	3	2
	PART B			
1a.	Vertices: S, A, B, C Edges: S→A (4), S→B (5), A→C (5), B→C (3), C→B (2) Number of vertices V=4 → relax edges V−1=3 times. Shortest distances from source S: • dist(S) = 0 • dist(A) = 4 • dist(B) = 5 • dist(C) = 8 No Negative cycles Bellman–Ford runs in O(V·E) time, because it relaxes all E edges V−1 times (plus one extra pass to check for negative cycles). For this graph V=4, E=5, so the runtime is O(4·5)=O(20) in constant-factor terms	4+2+1	3	2
1b.	DAG shortest path algorithms run faster due to: acyclicity guarantees a simple linear-time solution using topological sorting, no need for priority queues or repeated relaxations.	3	2	3
2.	Given matrices A1,A2,…, An where Ai is of dimension pi−1×pi, find the parenthesization of the product A1A2⋯ An that minimizes the total number of scalar multiplications. Complexity • Time: trying all l (≈n), all i (≈n), and all k (≈n) → O(n3). • Space: storing m and s → O(n2)	5+5	1	4
3.	Maximum flow: 13 Flow paths used: 1. S→A→B→T : 4 units 2. S→C→D→T : 9 units	3+5+2	3	3

				
4a.	Topological Order: A, B, C, D, E Shortest Paths from A: A:0, B:2, C:5, D:3, E: 6  • Shortest path from A → B is 2 • Shortest path from A → C is 5 • Shortest path from A → D is 3 (via B) • Shortest path from A → E is 6 (via D)	2+5	3	3
4b.	The Ford–Fulkerson algorithm can fail to terminate with irrational capacities because it might keep adding increasingly smaller fractions of flow along augmenting paths, approaching the maximum flow asymptotically but never exactly reaching it in a finite number of steps.	3	2	1
5.	Select Activity 1 (Finish = 4) → First activity. Next, choose the first activity whose start ≥ 4 → Activity 2 (Start = 3, no) → skip • Activity 3 (Start = 0 < 4) → skip • Activity 4 (Start = 5 ≥ 4) → select • Activity 5 (Start = 3 < 7) → skip • Activity 6 (Start = 5 < 7) → skip • Activity 7 (Start = 6 < 7) → skip • Activity 8 (Start = 8 ≥ 7) → select • Activity 9 (Start = 8 < 11) → skip Selected activities: 1, 4, 8	6+4	3	1

Course Outcomes: After completing the course, the students will be able to: -	
CO 1	Analyze various algorithms for their time and space complexity.
CO 2	Demonstrate a familiarity with major algorithms and data structures
CO 3	Apply appropriate design techniques for solving real world problems.
CO 4	Design and implement solutions using appropriate mathematical techniques.